YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 10 OF 10 · 40 MINUTES

Ice Engineers Expo 冰雪工程师成果展

Scientists share what they found, show their evidence, answer questions honestly and listen to feedback; that is how science gets checked.

- I can present my part of our investigation clearly in English, using key words and our numbers.
- I can ask or answer a question about evidence, and give kind, specific feedback.

Expo 成果展 · **presentation** 展示 · **audience** 观众 · **evidence** 证据 · **conclusion** 结论 · **feedback** 反馈 · **confident** 有把握的 · **scientist** 科学家

THE LESSON

Do Now (skills check)	15	Final skills check, same paper and conditions as Week 1; all papers collected at minute 15.
Hook	1	The Expo is open: order of talks and who asks whom; get talk frame, notes, card, slip ready.
Do (Expo talks)	15	2-minute talk per group at its poster; next group asks a card question; Expo notes, slips.
Think and Reflect	4	Pair talk: 'How do scientists find things out?'; feedback slips to groups; exit ticket.
Survey and certificates	5	Survey read aloud (3 min); thank the class; certificates or when and where; gallery walk.

Hands-on: No practical: 2-minute Expo talks at A3 posters (ice strips Plan A, dough strips Plan B); one question each.

Collect or assess: Skills check (column S, growth); survey (U to Z); strand 6 from talk record, strands 1 to 5 from posters (AC to AH).

SAFETY

- No practical hazards beyond normal classroom rules.
- Posters on walls or boards: keep the walkways in front of them clear of bags.

WATCH OUT

- Never shorten the skills check or skip the survey; if short of time, use 90-second talks and 3 questions.
- Fix talk order (next group asks; first asks last) and label each feedback slip with the group asked.

Language: Answer: 'Our evidence shows ___, but it does not show ___.' Band 1: buddy reads Chinese frame. Chinese counts for Core.

Next week: Last lesson: gallery walk, then store posters; mark and enter results; certificates, reports within a week.

PREP FOR ONE CLASS

Plan A: before the lesson about 66 min (21 could be done by someone else); after about 310 min (some times estimated)

Plan B: before the lesson about 66 min (21 could be done by someone else); after about 305 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Book printing with reprographics, incl. cards and certificates on card 5 min · teacher

THE WEEK BEFORE

- [ ] Print skills checks, surveys and rest of print job; cut surveys and slips 20 min · reprographics
- [ ] Plan room layout, sticky notes; book gallery walk and certificate ceremony 10 min · teacher

THE DAY BEFORE

- [ ] Fix talk order; fill Slides 3, 5, 10, talk record; print cards; label slips 15 min · teacher
- [ ] Only if given today: print certificates fixed by Week 9 levels (optional) optional 15 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Put posters up in talk order; set out papers, table kit, clipboard, timer 15 min · teacher

AFTER THE LESSON

- [ ] Collect and sort work; mark skills checks; enter scores and surveys 60 min · teacher
- [ ] Finalise strands 1 to 6; apply award rules; print and present certificates 60 min · teacher
- [ ] Read Expo notes and exit tickets; write and print 24 skills reports 90 min · teacher
- [ ] Return Week 1 and 10 skills checks after marking, if wanted (optional) optional 5 min · teacher
- [ ] Gallery walk signs up; then photograph, store posters; store and return kit 45 min · technician
- [ ] Programme summary; return or file logbooks; three lines for next year 45 min · teacher
- [ ] Plan A close-out: clear freezer shelves of leftover samples Plan A 10 min · technician
- [ ] Plan B close-out: throw away leftover dough and strips Plan B 5 min · technician

PRINT

- Skills check x24 (1 sheet each, double-sided, Week 1 master)
- Interest survey x24 (half-A4, 12 sheets, cut in half)
- Expo notes worksheet x24 (2 sheets each, double-sided)
- Peer feedback slips x24 (A6, 6 sheets, cut in four)
- Question cards x24 (2 sheets, A4 card if possible, cut)
- Gallery walk signs x2 (1 sheet each)
- Certificates x25 (A4 card: 24 awards + 1 Most Improved)
- Skills reports x24 (1 sheet each, after the lesson)
- Teacher copies: run sheet, talk record (2 sheets), kit card (A5)
- Skills check mark scheme, if Week 1 copy not kept (optional)

GET OR MAKE (NEW THIS LESSON)

- Pencils x2 and string x2 (gallery walk signs)
- Sticky notes x2 pads
- Clipboard x1
- Poster wall fixings (masking tape, putty or pins)
- Storage box labelled 'Ice Engineers' x1
- Guillotine or scissors x1
- Camera or phone x1 (school device)

Have ready from earlier weeks: A3 group posters x6 (Week 9), Group folders x6 with talk frames x24, Class box with logbooks x24, Role cards x6 sets (language buddy), Science dictionaries x24, Pencils x24, glue sticks (2 per group), Week 9 self-assessment slips x24, Week 1 skills checks x24 (optional)

Chinese: draft, needs native-speaker check